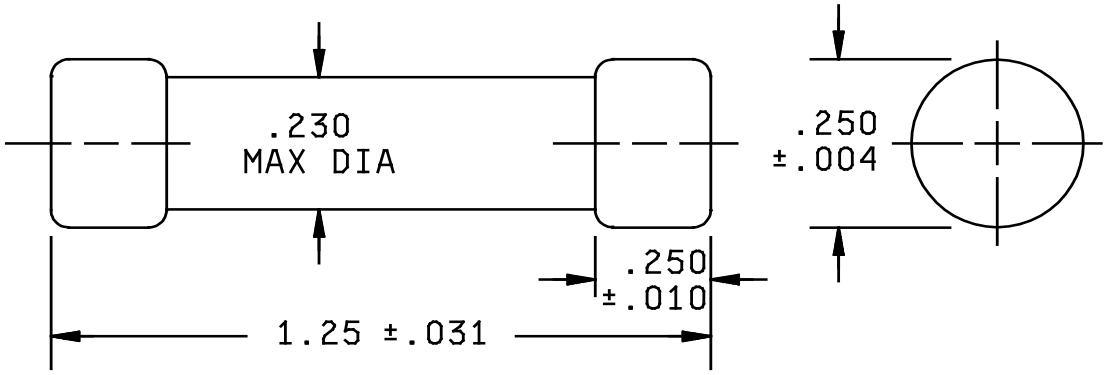
INCH-POUND
MIL-PRF-23419/9D
w/Amendment 1
21 April 2017
SUPERSEDING
MIL-PRF-23419/9D
17 April 2012

PERFORMANCE SPECIFICATION SHEET

# FUSE CARTRIDGE, INSTRUMENT TYPE, STYLE FM09, (NONINDICATING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-23419.




| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .004 | 0.10 | .031 | 0.79 | .250 | 6.35 |
| .010 | 0.25 | .230 | 5.84 | 1.25 | 31.75 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.

FIGURE 1. Style FM09 fuses.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:49Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Applicable fuseholders (or equivalent):

MIL-PRF-19207/1, Style FHL10U
MIL-PRF-19207/2, Style FHL11U
MIL-PRF-19207/8, Style FHL17G
MIL-PRF-19207/9, Style FHL18G
MIL-PRF-19207/11, Style FHL20G
MIL-PRF-19207/16, Style FHL26G
MIL-PRF-19207/17, Style FHL28WB
MIL-PRF-19207/18, Style FHL29G
MIL-PRF-19207/19, Style FHL30G
MIL-PRF-19207/21, Style FHL32W
MIL-PRF-19207/22, Style FHL33W

Terminals: Ferrule type.

Material: Brass, copper, or copper alloy.

Finish: Nickel or bright alloy plate, silver plated when specified.

Terminal strength: MIL-STD-202-211, test condition E, 2 pound-inch torque between ferrules and fuse body.

Body material: Plastic or ceramic tube.

Current rating: See table I.

Characteristic: See table I.

Voltage rating: See table I.

Current-carrying capacity for 20 amperes or less:

110 percent at room ambient temperature:

Characteristic A: 112.5 percent at -55$^{O}$C, 91.5 percent at +85$^{O}$C.

Characteristic B: 119 percent at -55$^{O}$C, 69.5 percent at +85$^{O}$C.

Current-carrying capacity for 25 amperes or 30 amperes:

100 percent with a maximum temperature rise of +90$^{O}$C for all temperatures between -55$^{O}$C and +85$^{O}$C.

Characteristic: See table I.

Overload interrupt: See table I (at all temperatures between -55$^{O}$C and +85$^{O}$C).

Resistance: See table I.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:49Z
Check the source to verify that this is the current version before use.

Short circuit interrupt: See table I.

Mechanical shock: Method I.

Thermal shock: MIL-STD-202-107, test condition A.

Vibration:

Characteristic A: All ratings, MIL-STD-202-204.

Characteristic B: Fuses rated 0 ampere through 8/10 ampere shall meet the requirements of MIL-STD-202-201

Part or Identifying Number (PIN): The PIN is derived from table I and shall be as shown in table IV.

TABLE I. PIN designation and electrical ratings.

| PIN designation | | | | Overload interrupt time 135 % of rating, maximum 1 hour 1/ | | AC short circuit Interrupt | | Cold resistance 2/ (ohms) | |
|---|---|---|---|---|---|---|---|---|---|
| Style | Characteristic | Maximum voltage | Current rating (amperes) | 200% of rating (seconds) | 300 % of rating (seconds) | 125 volts | 250 volts | Minimum | Maximum |
| FM09 | A | 250 V | 1/100 A | 0.0001 - 1.0 | 0.0001 - 0.003 | 10,000A | 35 A | 100.0 | 280.0 |
| | | 250 V | 1/32 A | 0.0008 - 1.0 | 0.0002 - 0.003 | | 35 A | 20.0 | 50.0 |
| | | 250 V | 1/16 A | 0.009 - 2.0 | 0.001 - 0.050 | | 35 A | 2.7 | 36.0 |
| | | 250 V | 1/8 A | 0.01 - 2.0 | 0.003 - 0.15 | | 35 A | 0.96 | 11.0 |
| | | 250 V | 1/4 A | 0.02 - 2.0 | 0.01 - 0.20 | | 35 A | 1.98 | 4.62 |
| | | 250 V | 3/8 A | 0.02 - 3.0 | 0.01 - 0.30 | | 35 A | .96 | 2.24 |
| | | 250 V | 1/2 A | 0.03 - 3.0 | 0.01 - 0.30 | | 35 A | .66 | 1.54 |
| | | 250 V | 3/4 A | 0.03 - 3.0 | 0.015 - 0.50 | | 35 A | .20 | .95 |
| | | 250 V | 1 A | 0.05 - 3.0 | 0.015 - 0.50 | | 35 A | .114 | .266 |
| | | 250 V | 1-1/4 A | 0.05 - 3.0 | 0.02 - 0.50 | | 35 A | .081 | .200 |
| | | 250 V | 1-1/2 A | 0.05 - 3.0 | 0.02 - 0.50 | | 35 A | .061 | .143 |
| | | 250 V | 2 A | 0.08 - 3.0 | 0.04 - 0.50 | | 35 A | .045 | .106 |
| | | 250 V | 3 A | 0.08 - 3.0 | 0.04 - 0.70 | | 35 A | .023 | .060 |
| | | 250 V | 4 A | 0.10 - 5.0 | 0.05 - 1.00 | | 35 A | .023 | .055 |
| | | 250 V | 5 A | .20 - 60.0 | 0.05 - 6.00 | | 100 A | .018 | .042 |
| | | 250 V | 6 A | .20 - 60.0 | 0.05 - 6.00 | | 100 A | .014 | .032 |
| | | 250 V | 8 A | .30 - 60.0 | 0.05 - 6.00 | | 100 A | .009 | .021 |
| | | 250 V | 10 A | .30 - 60.0 | 0.05 - 6.00 | | 100 A | .0068 | .0158 |
| | | 250 V | 12 A | .30 - 60.0 | 0.05 - 6.00 | | 200 A | .0045 | .0114 |
| | | 250 V | 15 A | 0.5 - 60.0 | 0.05 - 6.00 | | 200 A | .0027 | .0065 |
| | | 125 V | 20 A | 0.5 - 60.0 | 0.05 - 6.00 | | Not rated | .0016 | .0045 |
| | | 125 V | 25 A | 0.5 - 60.0 | 0.05 - 6.00 | | Not rated | .0013 | .0035 |
| | | 125 V | 30 A | 0.5 - 60.0 | 0.05 -6.00 | | Not rated | .0010 | .0027 |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:49Z
Check the source to verify that this is the current version before use.

TABLE I. PIN designation and electrical ratings - Continued.

| PIN designation | | | | Overload interrupt time 135 % of rating, maximum 1 hour 1/ | | AC short circuit interrupt | | Cold resistance 2/ (ohms) | |
|---|---|---|---|---|---|---|---|---|---|
| Style | Characteristic | Maximum voltage | Current rating (amperes) | 200% of rating (seconds) | 300 % of rating (seconds) | 125 volts | 250 volts | Minimum | Maximum |
| FM09 | B | 250 V | 1/100A | 4.0 - 250.0 | 2.0 - 40.0 | 10,000A | 35 A | 950.0 | 4620.0 |
| | | 250 V | 1/32 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | 100.0 | 462.0 |
| | | 250 V | 1/16 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | 49.0 | 120.0 |
| | | 250 V | 1/10 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | 14.0 | 17.5 |
| | | 250 V | 1/8 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | 9.0 | 11.0 |
| | | 250 V | 15/100 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | 6.9 | 8.0 |
| | | 250 V | 3/16 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | 4.0 | 5.5 |
| | | 250 V | 1/4 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | 3.18 | 10.5 |
| | | 250 V | 3/10 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | 2.22 | 7.10 |
| | | 250 V | 3/8 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | 1.5 | 4.77 |
| | | 250 V | 1/2 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | 1.2 | 2.83 |
| | | 250 V | 6/10 A | 4.0 - 0.0 | 2.0 - 25.0 | | 35 A | .60 | 2.10 |
| | | 250 V | 3/4 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | .45 | 1.43 |
| | | 250 V | 8/10 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | .35 | 1.26 |
| | | 250 V | 1 A | 4.0-80.0 | 2.0 - 25.0 | | 35 A | .21 | .64 |
| | | 250 V | 1-1/4 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | .16 | .38 |
| | | 250 V | 1-1/2 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | .102 | .340 |
| | | 250 V | 1-6/10 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | .099 | .270 |
| | | 250 V | 2 A | 4.0 - 80.0 | 2.0 - 25.0 | | 35 A | .072 | .180 |
| | | 250 V | 2-1/2 A | 10.0 - 250.0 | 2.0 - 25.0 | | 35 A | .052 | .154 |
| | | 250 V | 2-8/10 A | 10.0 - 250.0 | 2.0 - 25.0 | | 35 A | .032 | .090 |
| | | 250 V | 3 A | 10.0 - 250.0 | 2.0 - 25.0 | | 35 A | .026 | .090 |
| | | 250 V | 3-2/10 A | 10.0 - 250.0 | 2.0 - 25.0 | | 35 A | .025 | .090 |
| | | 125 V | 4 A | 10.0 - 250.0 | 2.0 - 25.0 | | Not rated | .0240 | .0560 |
| | | 125 V | 5 A | 10.0 - 250.0 | 0.05 - 10.0 | | Not rated | .0108 | .0600 |
| | | 125 V | 6-1/4 A | 10.0 - 250.0 | 0.05 - 10.0 | | Not rated | .0071 | .0450 |
| | | 125 V | 7 A | 10.0 - 250.0 | 0.05 - 10.0 | | Not rated | .0059 | .0300 |
| | | 125 V | 8 A | 10.0 - 250.0 | 0.05 - 10.0 | | Not rated | .0070 | .0200 |
| | | 125 V | 10 A | 10.0 - 250.0 | 0.05 - 10.0 | | Not rated | .0056 | .0140 |
| | | 125 V | 12 A | 10.0 - 250.0 | 0.05 - 10.0 | | Not rated | .0036 | .0110 |
| | | 125 V | 15 A | 10.0 - 250.0 | 0.05 - 10.0 | | Not rated | .0017 | .0080 |
| | | 125 V | 20 A | 10.0 - 250.0 | 0.05 - 10.0 | | Not rated | .0012 | .0040 |
| | | 125 V | 25 A | 10.0 - 250.0 | 0.05 - 10.0 | | Not rated | .0009 | .0030 |
| | | 125 V | 30 A | 10.0 - 250.0 | 0.05 - 10.0 | | Not rated | .0006 | .0022 |

1/ The overload interrupt time ranges, as shown, cover the temperature range from -55°C to +85°C. For characteristic B, the overload interrupt time at room ambient temperature shall be 12 seconds minimum at 200 percent of rating.
2/ Cold resistance is measured at 10 percent or less of rated current.
3/ Add “S” suffix to the end of the PIN designation if optional silver plating is required.

VERIFICATION:

Inspection routine The number of group II samples for qualification and subgroup 2 samples for group C inspection shall be as shown in table II.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:49Z
Check the source to verify that this is the current version before use.

TABLE II. Group II and subgroup 2 samples. 1/

| Inspection | Number of sample fuses |
|---|---|
| Group II and subgroup 2 | 20 |
| Terminal strength | 20 |
| Overload interrupt | |
| 135% at room ambient | 4 |
| 200 percent at -55°C | 4 |
| 200 percent at +85°C | 4 |
| 300 percent at -55°C | 4 |
| 300 percent at +85°C | 4 |

1/ For characteristic B fuses only, four additional fuses shall be tested at 200 percent of rating at room ambient temperature.

The number of group III samples for qualification and subgroup 1 for group C inspection, shall consist of four fuses of each of the styles shown in table III. Short circuit interrupt tests shall be conducted at the alternating current voltage indicated, and the fuses shall be capable of withstanding a minimum short circuit current as shown in table III.

TABLE III. Short circuit tests.

| Fuse style | Current rating | Short circuit current | AC type voltage |
|---|---|---|---|
| FM09A | 4 A | 35 A | 250 |
| FM09A | 10 A | 100 A | 250 |
| FM09A | 15 A | 200 A | 250 |
| FM09A | 30 A | 10,000 A | 125 |
| FM09B | 3 2/10 A | 35 A | 250 |
| FM09B | 30 A | 10,000 A | 125 |

Fuse selection: The following steps should apply in the selection of a fuse for any application:

Step 1: Select a fuse with a voltage rating equal to or in excess of the circuit voltage.

Step 2: The short circuit interrupting rating of the fuse should be checked to determine that it is adequate to protect against the short circuit currents that the particular equipment or circuit can be exposed to.

Step 3: The ambient temperature in which the fuse is to be used should be compared to the curves shown on figure 1 or figure 2 to determine the effect the ambient will have on the current rating. For example, if a characteristic A fuse is to be used in a 60°C ambient, it will only carry 95 percent of its nominal rating. A fuse which is used in a -20°C ambient will carry 107 percent of its nominal rating.

Step 4: A fuse rating should be selected on the basis of equipment ampere load with compensation made for ambient temperature (see figure 2 and figure 3). Characteristic A and characteristic B fuses should not be loaded to more than 80 percent of the fuse rating (or 80 percent of the adjusted rating based on ambient temperature consideration) to avoid nuisance openings caused by inrush currents and other harmless overloads. Time lag (characteristic B) fuses can be loaded to a higher level because of the ability to override harmless overloads, transients, and inrush currents of short duration.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:49Z
Check the source to verify that this is the current version before use.

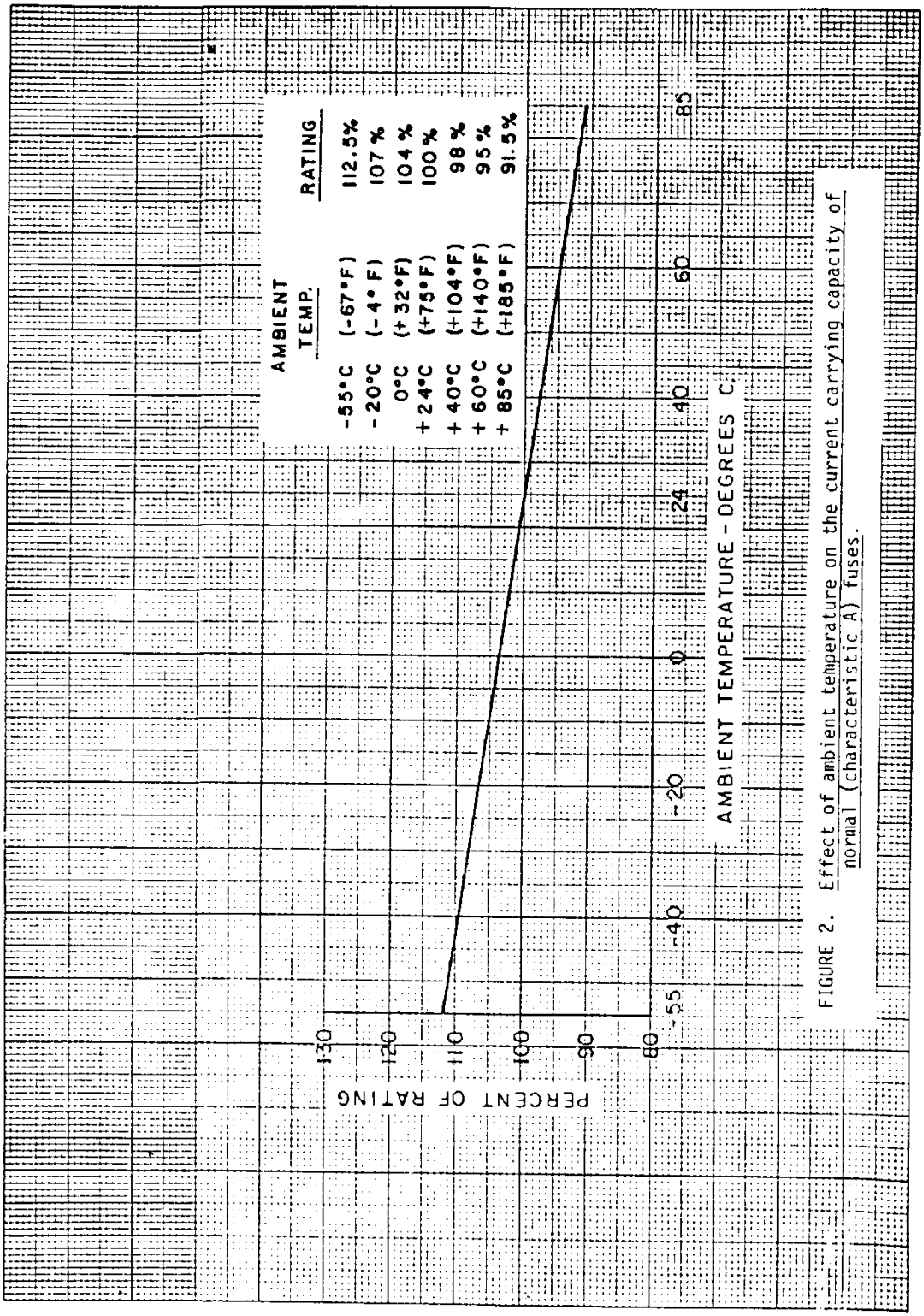



FIGURE 2. Effect of ambient temperature on the current carrying capacity of normal (characteristic A) fuses.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:49Z
Check the source to verify that this is the current version before use.

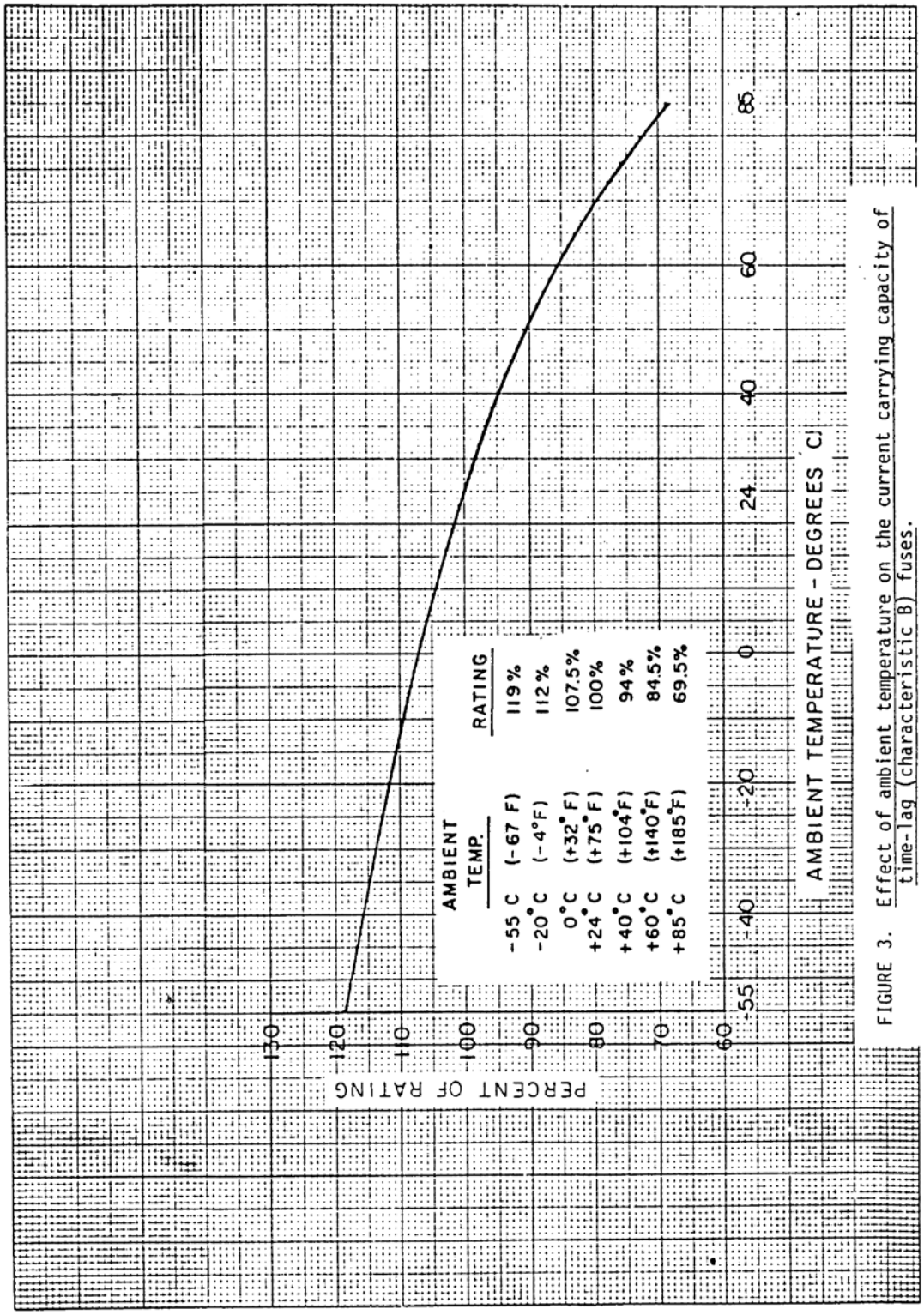



FIGURE 3. Effect of ambient temperature on the current carrying capacity of time-lag (characteristic B) fuses.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:49Z
Check the source to verify that this is the current version before use.

Cross reference: For applicable cross reference, see table IV. The existing stocks of superseded items may be used. When exhausted, the superseding parts shall be used.

TABLE IV. Cross-reference.

| Superseding PIN MIL-PRF-23419/9 | Superseded number MIL-F-23419/3 | Superseding PIN MIL-PRF-23419/9 | Superseded number MIL-F-23419/6 |
|---|---|---|---|
| FM09A250V1/100A | FM03A 250V 1/100A | FM09B250V1/100A | FM06-250V 1/100A |
| FM09A250V1/32A | FM03A 250V 1/32A | FM09B250V1/32A | FM06-250V 1/32A |
| FM09A250V1/16A | FM03A 250V 1/16A | FM09B250V1/16A | FM06-250V 1/16A |
| FM09A250V1/8A | FM03A 250V 1/8A | FM09B250V1/10A | FM06-250V 1/10A |
| FM09A250V1/4A | FM03A 250V 1/4A | FM09B250V1/8A | FM06-250V 1/8A |
| FM09A250V3/8A | FM03A 250V 3/8A | FM09B250V15/100A | FM06-250V 15/100A |
| FM09A250V1/2A | FM03A 250V 1/2A | FM09B250V3/16A | FM06-250V 3/16A |
| FM09A250V3/4A | FM03A 250V 3/4A | FM09B250V2/10A | FM06-250V 2/10A |
| FM09A250V1A | FM03A 250V 1A | FM09B250V1/4A | FM06-250V 1/4A |
| FM09A250V1-1/2A | FM03A 250V 1-1/2A | FM09B250V3/10A | FM06-250V 3/10A |
| FM09A250V2A | FM03A 250V 2A | FM09B250V3/8A | FM06-250V 3/8A |
| FM09A250V3A | FM03A 250V 3A | FM09B250V4/10A | FM06-250V 4/10A |
| FM09A250V5A | FM03A 250V 4A | FM09B250V1/2A | FM06-250V 1/2A |
| FM09A250V6A | FM03A 250V 5A | FM09B250V6/10A | FM06-250V 6/10A |
| FM09A250V8A | FM03A 250V 6A | FM09B250V7/10A | FM06-250V 7/10A |
| FM09A250V10A | FM03A 250V 8A | FM09B250V3/4A | FM06-250V 3/4A |
| FM09A250V12A | FM03A 250V 10A | FM09B250V1A | FM06-250V 1A |
| FM09A250V15A | FM03A 250V 12A | FM09B250V1-1/4A | FM06-250V 1-1/4A |
| FM09A125V20A | FM03A 125V 15A | FM09B250V1-1/2A | FM06-250V 1-1/2A |
| FM09A125V25A | FM03A 125V 20A | FM09B250V1-6/10A | FM06-250V 1-6/10A |
| FM09A125V30A | FM03A 125V 25A | FM09B250V2A | FM06-250V 2A |
| N/A | N/A | FM09B250V2-1/2A | FM06-125V 2-1/2A |
| N/A | N/A | FM09B250V2-8/10A | FM06-125V 2-8/10A |
| N/A | N/A | FM09B250V3A | FM06-125V 3A |
| N/A | N/A | FM09B250V3-2/10A | FM06-125V 3-2/10A |
| N/A | N/A | FM09B250V4A | FM06-125V 4A |
| N/A | N/A | FM09B250V5A | FM06-125V 5A |
| N/A | N/A | FM09B250V6-1/4A | FM06-125V 6-1/4A |
| N/A | N/A | FM09B250V7A | FM06-125V 7A |
| N/A | N/A | FM09B250V8A | FM06-125V 8A |
| N/A | N/A | FM09B250V10A | FM06-125V 10A |
| N/A | N/A | FM09B250V12A | FM06-125V 12A |
| N/A | N/A | FM09B250V15A | FM06-125V 15A |
| N/A | N/A | FM09B250V20A | FM06-125V 20A |
| N/A | N/A | FM09B250V25A | FM06-125V 25A |
| N/A | N/A | FM09B250V30A | FM06-125V 30A |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:49Z
Check the source to verify that this is the current version before use.

Referenced documents. In addition to MIL-PRF-23419, this document references the following:

MIL-STD-202-107
MIL-STD-202-201
MIL-STD-202-204
MIL-STD-202-211

MIL-PRF-19207/1
MIL-PRF-19207/2
MIL-PRF-19207/8
MIL-PRF-19207/9
MIL-PRF-19207/11
MIL-PRF-19207/16
MIL-PRF-19207/17
MIL-PRF-19207/18
MIL-PRF-19207/19
MIL-PRF-19207/21
MIL-PRF-19207/22

The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5920-2017-023)

Review activities:
Army - AR, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99
NSA - NA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:49Z
Check the source to verify that this is the current version before use.